

CSMFAB000000000017 Aegis Home_ Dielectric Rope Fabrication

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** CSMFAB000000000017 Aegis Home_ Dielectric Rope Fabrication V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
 3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete
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V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
YInMn Blue spectral coating	V1.0: YInMn + 2 mol% Ta ₂ O ₅ ; SRI 115; NIR 85%	V2.0: CsPbBr ₃ QD co-deposit; SRI ~130; UV reflectance +31%	UV band (280-400 nm) protection gap closed; QD stable at 1200°C co-fire	Solar Energy Mater. Sol. Cells 258 (2025) 112456
Composite matrix — recyclability	V1.0: Thermoset epoxy; irreversible crosslink; landfill end-of-life	V2.0: Elium thermoplastic acrylic; thermal solvolysis 350°C; 95% fiber + 100% monomer recovery	Thermoset blocks Phoenix Protocol circular economy mandate	Arkema Elium TDS 2025; Composites Part A 2024

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
YInMn Blue spectral coating	UV band protection gap; accelerated UV-induced substrate polymer degradation	MEDIUM
Composite matrix — recyclability	No end-of-life recovery path; contradicts Phoenix Protocol; Phoenix Protocol breach	HIGH

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMFAB000000000017 Aegis Home_ Dielectric Rope Fabrication V1.0 archived; V2.0 is the active specification as of June 2026.